

```
In [1]: import os
        from dotenv import load_dotenv
        load_dotenv()
        OPENAI_API_KEY = os.getenv("OPENAI_API_KEY")
```

```
In [2]: !pip install pydantic
```

```
Requirement already satisfied: pydantic in c:\programdata\anaconda3\lib\site-packages (2.10.3)
Requirement already satisfied: annotated-types>=0.6.0 in c:\programdata\anaconda3\lib\site-packages (from pydantic) (0.6.0)
Requirement already satisfied: pydantic-core==2.27.1 in c:\programdata\anaconda3\lib\site-packages (from pydantic) (2.27.1)
Requirement already satisfied: typing-extensions>=4.12.2 in c:\programdata\anaconda3\lib\site-packages (from pydantic) (4.15.0)
```

```
In [3]: !pip install langchain openai langchain-openai langchain-community
```

Requirement already satisfied: langchain in c:\programdata\anaconda3\lib\site-packages (1.2.3)
Requirement already satisfied: openai in c:\programdata\anaconda3\lib\site-packages (2.15.0)
Requirement already satisfied: langchain-openai in c:\programdata\anaconda3\lib\site-packages (1.1.7)
Requirement already satisfied: langchain-community in c:\programdata\anaconda3\lib\site-packages (0.4.1)
Requirement already satisfied: langchain-core<2.0.0,>=1.2.1 in c:\programdata\anaconda3\lib\site-packages (from langchain) (1.2.7)
Requirement already satisfied: langgraph<1.1.0,>=1.0.2 in c:\programdata\anaconda3\lib\site-packages (from langchain) (1.0.6)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in c:\programdata\anaconda3\lib\site-packages (from langchain) (2.10.3)
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in c:\programdata\anaconda3\lib\site-packages (from langchain-core<2.0.0,>=1.2.1->langchain) (1.33)
Requirement already satisfied: langsmith<1.0.0,>=0.3.45 in c:\programdata\anaconda3\lib\site-packages (from langchain-core<2.0.0,>=1.2.1->langchain) (0.6.2)
Requirement already satisfied: packaging<26.0.0,>=23.2.0 in c:\programdata\anaconda3\lib\site-packages (from langchain-core<2.0.0,>=1.2.1->langchain) (24.2)
Requirement already satisfied: pyyaml<7.0.0,>=5.3.0 in c:\programdata\anaconda3\lib\site-packages (from langchain-core<2.0.0,>=1.2.1->langchain) (6.0.2)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in c:\programdata\anaconda3\lib\site-packages (from langchain-core<2.0.0,>=1.2.1->langchain) (9.0.0)
Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in c:\programdata\anaconda3\lib\site-packages (from langchain-core<2.0.0,>=1.2.1->langchain) (4.15.0)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in c:\programdata\anaconda3\lib\site-packages (from langchain-core<2.0.0,>=1.2.1->langchain) (0.13.0)
Requirement already satisfied: jsonpointer>=1.9 in c:\programdata\anaconda3\lib\site-packages (from jsonpatch<2.0.0,>=1.33.0->langchain-core<2.0.0,>=1.2.1->langchain) (2.1)
Requirement already satisfied: langgraph-checkpoint<5.0.0,>=2.1.0 in c:\programdata\anaconda3\lib\site-packages (from langgraph<1.1.0,>=1.0.2->langchain) (4.0.0)
Requirement already satisfied: langgraph-prebuilt<1.1.0,>=1.0.2 in c:\programdata\anaconda3\lib\site-packages (from langgraph<1.1.0,>=1.0.2->langchain) (1.0.6)
Requirement already satisfied: langgraph-sdk<0.4.0,>=0.3.0 in c:\programdata\anaconda3\lib\site-packages (from langgraph<1.1.0,>=1.0.2->langchain) (0.3.3)
Requirement already satisfied: xxhash>=3.5.0 in c:\programdata\anaconda3\lib\site-packages (from langgraph<1.1.0,>=1.0.2->langchain) (3.6.0)
Requirement already satisfied: ormsgpack>=1.12.0 in c:\programdata\anaconda3\lib\site-packages (from langgraph-checkpoint<5.0.0,>=2.1.0->langgraph<1.1.0,>=1.0.2->langchain) (1.12.1)
Requirement already satisfied: httpx>=0.25.2 in c:\programdata\anaconda3\lib\site-packages (from langgraph-sdk<0.4.0,>=0.3.0->langgraph<1.1.0,>=1.0.2->langchain) (0.28.1)
Requirement already satisfied: orjson>=3.10.1 in c:\programdata\anaconda3\lib\site-packages (from langgraph-sdk<0.4.0,>=0.3.0->langgraph<1.1.0,>=1.0.2->langchain) (3.11.5)
Requirement already satisfied: requests-toolbelt>=1.0.0 in c:\programdata\anaconda3\lib\site-packages (from langsmith<1.0.0,>=0.3.45->langchain-core<2.0.0,>=1.2.1->langchain) (1.0.0)
Requirement already satisfied: requests>=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from langsmith<1.0.0,>=0.3.45->la

ngchain-core<2.0.0,>=1.2.1->langchain) (2.32.5)
Requirement already satisfied: zstandard>=0.23.0 in c:\programdata\anaconda3\lib\site-packages (from langsmith<1.0.0,>=0.3.45->langchain-core<2.0.0,>=1.2.1->langchain) (0.23.0)
Requirement already satisfied: anyio in c:\programdata\anaconda3\lib\site-packages (from httpx>=0.25.2->langgraph-sdk<0.4.0,>=0.3.0->langgraph<1.1.0,>=1.0.2->langchain) (4.7.0)
Requirement already satisfied: certifi in c:\programdata\anaconda3\lib\site-packages (from httpx>=0.25.2->langgraph-sdk<0.4.0,>=0.3.0->langgraph<1.1.0,>=1.0.2->langchain) (2025.10.5)
Requirement already satisfied: httpcore==1.* in c:\programdata\anaconda3\lib\site-packages (from httpx>=0.25.2->langgraph-sdk<0.4.0,>=0.3.0->langgraph<1.1.0,>=1.0.2->langchain) (1.0.9)
Requirement already satisfied: idna in c:\programdata\anaconda3\lib\site-packages (from httpx>=0.25.2->langgraph-sdk<0.4.0,>=0.3.0->langgraph<1.1.0,>=1.0.2->langchain) (3.7)
Requirement already satisfied: h11>=0.16 in c:\programdata\anaconda3\lib\site-packages (from httpcore==1.*->httpx>=0.25.2->langgraph-sdk<0.4.0,>=0.3.0->langgraph<1.1.0,>=1.0.2->langchain) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in c:\programdata\anaconda3\lib\site-packages (from pydantic<3.0.0,>=2.7.4->langchain) (0.6.0)
Requirement already satisfied: pydantic-core==2.27.1 in c:\programdata\anaconda3\lib\site-packages (from pydantic<3.0.0,>=2.7.4->langchain) (2.27.1)
Requirement already satisfied: distro<2,>=1.7.0 in c:\programdata\anaconda3\lib\site-packages (from openai) (1.9.0)
Requirement already satisfied: jiter<1,>=0.10.0 in c:\programdata\anaconda3\lib\site-packages (from openai) (0.12.0)
Requirement already satisfied: sniffio in c:\programdata\anaconda3\lib\site-packages (from openai) (1.3.0)
Requirement already satisfied: tqdm>4 in c:\programdata\anaconda3\lib\site-packages (from openai) (4.67.1)
Requirement already satisfied: tiktoken<1.0.0,>=0.7.0 in c:\programdata\anaconda3\lib\site-packages (from langchain-openai) (0.12.0)
Requirement already satisfied: regex>=2022.1.18 in c:\programdata\anaconda3\lib\site-packages (from tiktoken<1.0.0,>=0.7.0->langchain-openai) (2024.11.6)
Requirement already satisfied: langchain-classic<2.0.0,>=1.0.0 in c:\programdata\anaconda3\lib\site-packages (from langchain-community) (1.0.1)
Requirement already satisfied: SQLAlchemy<3.0.0,>=1.4.0 in c:\programdata\anaconda3\lib\site-packages (from langchain-community) (2.0.39)
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in c:\programdata\anaconda3\lib\site-packages (from langchain-community) (3.11.10)
Requirement already satisfied: dataclasses-json<0.7.0,>=0.6.7 in c:\programdata\anaconda3\lib\site-packages (from langchain-community) (0.6.7)
Requirement already satisfied: pydantic-settings<3.0.0,>=2.10.1 in c:\programdata\anaconda3\lib\site-packages (from langchain-community) (2.12.0)
Requirement already satisfied: httpx-sse<1.0.0,>=0.4.0 in c:\programdata\anaconda3\lib\site-packages (from langchain-community) (0.4.3)
Requirement already satisfied: numpy>=2.1.0 in c:\programdata\anaconda3\lib\site-packages (from langchain-community) (2.1.3)
Requirement already satisfied: aiohappyeyeballs>=2.3.0 in c:\programdata\anaconda3\lib\site-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (2.4.4)
Requirement already satisfied: aiosignal>=1.1.2 in c:\programdata\anaconda3\lib\site-packages (from aiohttp<4.0.0,>=3.8.3->lang

```

chain-community) (1.2.0)
Requirement already satisfied: attrs>=17.3.0 in c:\programdata\anaconda3\lib\site-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (24.3.0)
Requirement already satisfied: frozenlist>=1.1.1 in c:\programdata\anaconda3\lib\site-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (1.5.0)
Requirement already satisfied: multidict<7.0,>=4.5 in c:\programdata\anaconda3\lib\site-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (6.1.0)
Requirement already satisfied: propcache>=0.2.0 in c:\programdata\anaconda3\lib\site-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (0.3.1)
Requirement already satisfied: yarll<2.0,>=1.17.0 in c:\programdata\anaconda3\lib\site-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (1.18.0)
Requirement already satisfied: marshmallow<4.0.0,>=3.18.0 in c:\programdata\anaconda3\lib\site-packages (from dataclasses-json<0.7.0,>=0.6.7->langchain-community) (3.26.2)
Requirement already satisfied: typing-inspect<1,>=0.4.0 in c:\programdata\anaconda3\lib\site-packages (from dataclasses-json<0.7.0,>=0.6.7->langchain-community) (0.9.0)
Requirement already satisfied: langchain-text-splitters<2.0.0,>=1.1.0 in c:\programdata\anaconda3\lib\site-packages (from langchain-classic<2.0.0,>=1.0.0->langchain-community) (1.1.0)
Requirement already satisfied: python-dotenv>=0.21.0 in c:\programdata\anaconda3\lib\site-packages (from pydantic-settings<3.0.0,>=2.10.1->langchain-community) (1.1.0)
Requirement already satisfied: typing-inspection>=0.4.0 in c:\programdata\anaconda3\lib\site-packages (from pydantic-settings<3.0.0,>=2.10.1->langchain-community) (0.4.2)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\programdata\anaconda3\lib\site-packages (from requests>=2.0.0->langsmith<1.0.0,>=0.3.45->langchain-core<2.0.0,>=1.2.1->langchain) (3.3.2)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\programdata\anaconda3\lib\site-packages (from requests>=2.0.0->langsmith<1.0.0,>=0.3.45->langchain-core<2.0.0,>=1.2.1->langchain) (2.3.0)
Requirement already satisfied: greenlet!=0.4.17 in c:\programdata\anaconda3\lib\site-packages (from SQLAlchemychemy<3.0.0,>=1.4.0->langchain-community) (3.1.1)
Requirement already satisfied: mypy-extensions>=0.3.0 in c:\programdata\anaconda3\lib\site-packages (from typing-inspect<1,>=0.4.0->dataclasses-json<0.7.0,>=0.6.7->langchain-community) (1.0.0)
Requirement already satisfied: colorama in c:\programdata\anaconda3\lib\site-packages (from tqdm>4->openai) (0.4.6)

```

5-1 Data connection

Document loaders

```
In [4]: !pip install GitPython
```

Requirement already satisfied: GitPython in c:\programdata\anaconda3\lib\site-packages (3.1.43)
Requirement already satisfied: gitdb<5,>=4.0.1 in c:\programdata\anaconda3\lib\site-packages (from GitPython) (4.0.7)
Requirement already satisfied: smmap<5,>=3.0.1 in c:\programdata\anaconda3\lib\site-packages (from gitdb<5,>=4.0.1->GitPython) (4.0.0)

```
In [5]: from langchain_classic.document_loaders import GitLoader

def file_filter(file_path):
    return file_path.endswith(".md")

loader = GitLoader(
    clone_url="https://github.com/langchain-ai/langchain",
    repo_path="./langchain",
    branch="master",
    file_filter=file_filter,
)

raw_docs = loader.load()
print(len(raw_docs))
```

36

Document transformers

```
In [6]: from langchain_classic.text_splitter import CharacterTextSplitter

text_splitter = CharacterTextSplitter(chunk_size=1000, chunk_overlap=0)

docs = text_splitter.split_documents(raw_docs)
print(len(docs))
```

Created a chunk of size 1367, which is longer than the specified 1000
Created a chunk of size 1455, which is longer than the specified 1000
Created a chunk of size 1299, which is longer than the specified 1000

96

Text embedding models

```
In [7]: from langchain_openai import OpenAIEmbeddings
```

```
embeddings = OpenAIEmbeddings()
```

```
In [8]: !pip install tiktoken
```

Requirement already satisfied: tiktoken in c:\programdata\anaconda3\lib\site-packages (0.12.0)

Requirement already satisfied: regex>=2022.1.18 in c:\programdata\anaconda3\lib\site-packages (from tiktoken) (2024.11.6)

Requirement already satisfied: requests>=2.26.0 in c:\programdata\anaconda3\lib\site-packages (from tiktoken) (2.32.5)

Requirement already satisfied: charset_normalizer<4,>=2 in c:\programdata\anaconda3\lib\site-packages (from requests>=2.26.0->tiktoken) (3.3.2)

Requirement already satisfied: idna<4,>=2.5 in c:\programdata\anaconda3\lib\site-packages (from requests>=2.26.0->tiktoken) (3.7)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\programdata\anaconda3\lib\site-packages (from requests>=2.26.0->tiktoken) (2.3.0)

Requirement already satisfied: certifi>=2017.4.17 in c:\programdata\anaconda3\lib\site-packages (from requests>=2.26.0->tiktoken) (2025.10.5)

```
In [16]: query = "AWS의 S3에서 데이터를 읽기 위한 DocumentLoader가 있나요?"
```

```
vector = embeddings.embed_query(query)
print(len(vector))
print(vector[:100])
```

1536

```
[-0.01556143257766962, -0.012867593206465244, 0.021737877279520035, -0.01614966429769993, -0.024117546156048775, 0.045186977833  
509445, -0.019705800339579582, -0.0077606625854969025, -0.0040641543455421925, 0.022138945758342743, 0.01052134670317173, 0.015  
49458783119917, 0.0010402697371318936, -0.018796712160110474, -0.007393017411231995, -0.002800790360197425, 0.0177271980792284,  
-0.00953204557299614, -0.009244613349437714, -0.03665760159492493, 0.012967860326170921, -0.015628276392817497, -0.005621634423  
732758, 0.0017429740400984883, -0.0026554034557193518, 0.009017341770231724, 0.007687133736908436, -0.04016026109457016, 0.0072  
19220977276564, 0.0026470478624105453, 0.0526735782623291, -0.013041389174759388, -0.0068415491841733456, -0.02272717840969562  
5, -0.010026696138083935, -0.014705820940434933, 0.00838231761008501, -0.0006972731789574027, -0.006173102650791407, -0.0215239  
7483587265, 0.002012023702263832, -0.006493956781923771, -0.0006116284639574587, -0.04347575455904007, 0.002316166879609227, 0.  
02005339227616787, 0.002648718887940049, -0.04251319169998169, -0.007981250062584877, 0.01530742272734642, 0.00345252570696175  
1, 0.02478599362075329, -0.015588169917464256, -0.0372190959751606, -0.026858177036046982, 0.012159040197730064, -0.01664431579  
4110298, 0.0018749921582639217, -0.007339541334658861, -0.027834108099341393, 0.007573497947305441, 0.033235155045986176, -0.01  
822184957563877, -0.005220566410571337, 0.002093908376991749, -0.007693818304687738, -0.010327496565878391, 0.00346589460968971  
25, 0.009565467946231365, -0.011397011578083038, 0.005434469319880009, 0.03018704056739807, 0.004468564409762621, -0.0225533824  
4140148, 0.03572177514433861, -0.005076850298792124, -0.009812792763113976, -0.01508015114814043, -0.031818047165870667, -0.004  
197843372821808, 0.012312782928347588, -0.01720581017434597, 0.00178809417411685, -0.005912408232688904, 0.003943833988159895,  
0.020441090688109398, 0.015213840641081333, 0.02868971973657608, -0.008455847389996052, -0.008495953865349293, -0.0096389967948  
19832, -0.0004908903501927853, 0.01054139994084835, 0.015240577980875969, 0.020815420895814896, 0.02199188619852066, -0.0139036  
84914112091, 0.027138924226164818, -0.01663094572722912, -0.040427640080451965]
```

Vector stores

```
In [10]: !pip install chromadb
```

Requirement already satisfied: chromadb in c:\programdata\anaconda3\lib\site-packages (1.4.0)
Requirement already satisfied: build>=1.0.3 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (1.4.0)
Requirement already satisfied: pydantic>=1.9 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (2.10.3)
Requirement already satisfied: pybase64>=1.4.1 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (1.4.3)
Requirement already satisfied: uvicorn>=0.18.3 in c:\programdata\anaconda3\lib\site-packages (from uvicorn[standard]>=0.18.3->chromadb) (0.40.0)
Requirement already satisfied: numpy>=1.22.5 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (2.1.3)
Requirement already satisfied: posthog<6.0.0,>=2.4.0 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (5.4.0)
Requirement already satisfied: typing-extensions>=4.5.0 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (4.15.0)
Requirement already satisfied: onnxruntime>=1.14.1 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (1.23.2)
Requirement already satisfied: opentelemetry-api>=1.2.0 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (1.39.1)
Requirement already satisfied: opentelemetry-exporter-otlp-proto-grpc>=1.2.0 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (1.39.1)
Requirement already satisfied: opentelemetry-sdk>=1.2.0 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (1.39.1)
Requirement already satisfied: tokenizers>=0.13.2 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (0.22.2)
Requirement already satisfied: pypika>=0.48.9 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (0.48.9)
Requirement already satisfied: tqdm>=4.65.0 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (4.67.1)
Requirement already satisfied: overrides>=7.3.1 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (7.4.0)
Requirement already satisfied: importlib-resources in c:\programdata\anaconda3\lib\site-packages (from chromadb) (6.5.2)
Requirement already satisfied: grpcio>=1.58.0 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (1.76.0)
Requirement already satisfied: bcrypt>=4.0.1 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (4.3.0)
Requirement already satisfied: typer>=0.9.0 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (0.9.0)
Requirement already satisfied: kubernetes>=28.1.0 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (34.1.0)
Requirement already satisfied: tenacity>=8.2.3 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (9.0.0)
Requirement already satisfied: pyyaml>=6.0.0 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (6.0.2)
Requirement already satisfied: mmh3>=4.0.1 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (5.2.0)
Requirement already satisfied: orjson>=3.9.12 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (3.11.5)
Requirement already satisfied: httpx>=0.27.0 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (0.28.1)
Requirement already satisfied: rich>=10.11.0 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (13.9.4)
Requirement already satisfied: jsonschema>=4.19.0 in c:\programdata\anaconda3\lib\site-packages (from chromadb) (4.23.0)
Requirement already satisfied: requests<3.0,>=2.7 in c:\programdata\anaconda3\lib\site-packages (from posthog<6.0.0,>=2.4.0->chromadb) (2.32.5)
Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-packages (from posthog<6.0.0,>=2.4.0->chromadb) (1.17.0)
Requirement already satisfied: python-dateutil>=2.2 in c:\programdata\anaconda3\lib\site-packages (from posthog<6.0.0,>=2.4.0->chromadb) (2.9.0.post0)
Requirement already satisfied: backoff>=1.10.0 in c:\programdata\anaconda3\lib\site-packages (from posthog<6.0.0,>=2.4.0->chromadb) (2.2.1)
Requirement already satisfied: distro>=1.5.0 in c:\programdata\anaconda3\lib\site-packages (from posthog<6.0.0,>=2.4.0->chromadb) (1.9.0)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\programdata\anaconda3\lib\site-packages (from requests<3.0,>=2.7-

>posthog<6.0.0,>=2.4.0->chromadb) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in c:\programdata\anaconda3\lib\site-packages (from requests<3.0,>=2.7->posthog<6.0.0,>=2.4.0->chromadb) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\programdata\anaconda3\lib\site-packages (from requests<3.0,>=2.7->posthog<6.0.0,>=2.4.0->chromadb) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\programdata\anaconda3\lib\site-packages (from requests<3.0,>=2.7->posthog<6.0.0,>=2.4.0->chromadb) (2025.10.5)
Requirement already satisfied: packaging>=24.0 in c:\programdata\anaconda3\lib\site-packages (from build>=1.0.3->chromadb) (24.2)
Requirement already satisfied: pyproject_hooks in c:\programdata\anaconda3\lib\site-packages (from build>=1.0.3->chromadb) (1.2.0)
Requirement already satisfied: colorama in c:\programdata\anaconda3\lib\site-packages (from build>=1.0.3->chromadb) (0.4.6)
Requirement already satisfied: anyio in c:\programdata\anaconda3\lib\site-packages (from httpx>=0.27.0->chromadb) (4.7.0)
Requirement already satisfied: httpcore==1.* in c:\programdata\anaconda3\lib\site-packages (from httpx>=0.27.0->chromadb) (1.0.9)
Requirement already satisfied: h11>=0.16 in c:\programdata\anaconda3\lib\site-packages (from httpcore==1.*->httpx>=0.27.0->chromadb) (0.16.0)
Requirement already satisfied: attrs>=22.2.0 in c:\programdata\anaconda3\lib\site-packages (from jsonschema>=4.19.0->chromadb) (24.3.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in c:\programdata\anaconda3\lib\site-packages (from jsonschema>=4.19.0->chromadb) (2023.7.1)
Requirement already satisfied: referencing>=0.28.4 in c:\programdata\anaconda3\lib\site-packages (from jsonschema>=4.19.0->chromadb) (0.30.2)
Requirement already satisfied: rpds-py>=0.7.1 in c:\programdata\anaconda3\lib\site-packages (from jsonschema>=4.19.0->chromadb) (0.22.3)
Requirement already satisfied: google-auth>=1.0.1 in c:\programdata\anaconda3\lib\site-packages (from kubernetes>=28.1.0->chromadb) (2.47.0)
Requirement already satisfied: websocket-client!=0.40.0,!0.41.*,!0.42.*,>=0.32.0 in c:\programdata\anaconda3\lib\site-packages (from kubernetes>=28.1.0->chromadb) (1.8.0)
Requirement already satisfied: requests-oauthlib in c:\programdata\anaconda3\lib\site-packages (from kubernetes>=28.1.0->chromadb) (2.0.0)
Requirement already satisfied: durationpy>=0.7 in c:\programdata\anaconda3\lib\site-packages (from kubernetes>=28.1.0->chromadb) (0.10)
Requirement already satisfied: pyasn1-modules>=0.2.1 in c:\programdata\anaconda3\lib\site-packages (from google-auth>=1.0.1->kubernetes>=28.1.0->chromadb) (0.2.8)
Requirement already satisfied: rsa<5,>=3.1.4 in c:\programdata\anaconda3\lib\site-packages (from google-auth>=1.0.1->kubernetes>=28.1.0->chromadb) (4.9.1)
Requirement already satisfied: pyasn1>=0.1.3 in c:\programdata\anaconda3\lib\site-packages (from rsa<5,>=3.1.4->google-auth>=1.0.1->kubernetes>=28.1.0->chromadb) (0.4.8)
Requirement already satisfied: coloredlogs in c:\programdata\anaconda3\lib\site-packages (from onnxruntime>=1.14.1->chromadb) (15.0.1)

Requirement already satisfied: flatbuffers in c:\programdata\anaconda3\lib\site-packages (from onnxruntime>=1.14.1->chromadb) (25.12.19)

Requirement already satisfied: protobuf in c:\programdata\anaconda3\lib\site-packages (from onnxruntime>=1.14.1->chromadb) (5.29.3)

Requirement already satisfied: sympy in c:\programdata\anaconda3\lib\site-packages (from onnxruntime>=1.14.1->chromadb) (1.13.3)

Requirement already satisfied: importlib-metadata<8.8.0,>=6.0 in c:\programdata\anaconda3\lib\site-packages (from opentelemetry-api>=1.2.0->chromadb) (8.5.0)

Requirement already satisfied: zipp>=3.20 in c:\programdata\anaconda3\lib\site-packages (from importlib-metadata<8.8.0,>=6.0->opentelemetry-api>=1.2.0->chromadb) (3.21.0)

Requirement already satisfied: googleapis-common-protos~=1.57 in c:\programdata\anaconda3\lib\site-packages (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb) (1.72.0)

Requirement already satisfied: opentelemetry-exporter-otlp-proto-common==1.39.1 in c:\programdata\anaconda3\lib\site-packages (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb) (1.39.1)

Requirement already satisfied: opentelemetry-proto==1.39.1 in c:\programdata\anaconda3\lib\site-packages (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb) (1.39.1)

Requirement already satisfied: opentelemetry-semantic-conventions==0.60b1 in c:\programdata\anaconda3\lib\site-packages (from opentelemetry-sdk>=1.2.0->chromadb) (0.60b1)

Requirement already satisfied: annotated-types>=0.6.0 in c:\programdata\anaconda3\lib\site-packages (from pydantic>=1.9->chromadb) (0.6.0)

Requirement already satisfied: pydantic-core==2.27.1 in c:\programdata\anaconda3\lib\site-packages (from pydantic>=1.9->chromadb) (2.27.1)

Requirement already satisfied: markdown-it-py>=2.2.0 in c:\programdata\anaconda3\lib\site-packages (from rich>=10.11.0->chromadb) (2.2.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\programdata\anaconda3\lib\site-packages (from rich>=10.11.0->chromadb) (2.19.1)

Requirement already satisfied: mdurl~=0.1 in c:\programdata\anaconda3\lib\site-packages (from markdown-it-py>=2.2.0->rich>=10.11.0->chromadb) (0.1.0)

Requirement already satisfied: huggingface-hub<2.0,>=0.16.4 in c:\programdata\anaconda3\lib\site-packages (from tokenizers>=0.13.2->chromadb) (1.3.1)

Requirement already satisfied: filelock in c:\programdata\anaconda3\lib\site-packages (from huggingface-hub<2.0,>=0.16.4->tokenizers>=0.13.2->chromadb) (3.17.0)

Requirement already satisfied: fsspec>=2023.5.0 in c:\programdata\anaconda3\lib\site-packages (from huggingface-hub<2.0,>=0.16.4->tokenizers>=0.13.2->chromadb) (2025.3.2)

Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in c:\programdata\anaconda3\lib\site-packages (from huggingface-hub<2.0,>=0.16.4->tokenizers>=0.13.2->chromadb) (1.2.0)

Requirement already satisfied: shellingham in c:\programdata\anaconda3\lib\site-packages (from huggingface-hub<2.0,>=0.16.4->tokenizers>=0.13.2->chromadb) (1.5.0)

Requirement already satisfied: typer-slim in c:\programdata\anaconda3\lib\site-packages (from huggingface-hub<2.0,>=0.16.4->tokenizers>=0.13.2->chromadb) (0.21.1)

Requirement already satisfied: click<9.0.0,>=7.1.1 in c:\programdata\anaconda3\lib\site-packages (from typer>=0.9.0->chromadb)

(8.1.8)

Requirement already satisfied: httptools>=0.6.3 in c:\programdata\anaconda3\lib\site-packages (from uvicorn[standard]>=0.18.3->chromadb) (0.7.1)

Requirement already satisfied: python-dotenv>=0.13 in c:\programdata\anaconda3\lib\site-packages (from uvicorn[standard]>=0.18.3->chromadb) (1.1.0)

Requirement already satisfied: watchfiles>=0.13 in c:\programdata\anaconda3\lib\site-packages (from uvicorn[standard]>=0.18.3->chromadb) (1.1.1)

Requirement already satisfied: websockets>=10.4 in c:\programdata\anaconda3\lib\site-packages (from uvicorn[standard]>=0.18.3->chromadb) (16.0)

Requirement already satisfied: sniffio>=1.1 in c:\programdata\anaconda3\lib\site-packages (from anyio->httpx>=0.27.0->chromadb) (1.3.0)

Requirement already satisfied: humanfriendly>=9.1 in c:\programdata\anaconda3\lib\site-packages (from coloredlogs->onnxruntime>=1.14.1->chromadb) (10.0)

Requirement already satisfied: pyreadline3 in c:\programdata\anaconda3\lib\site-packages (from humanfriendly>=9.1->coloredlogs->onnxruntime>=1.14.1->chromadb) (3.5.4)

Requirement already satisfied: oauthlib>=3.0.0 in c:\programdata\anaconda3\lib\site-packages (from requests-oauthlib->kubernete>=28.1.0->chromadb) (3.3.1)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\programdata\anaconda3\lib\site-packages (from sympy->onnxruntime>=1.14.1->chromadb) (1.3.0)

```
In [11]: from langchain_classic.vectorstores import Chroma
```

```
db = Chroma.from_documents(docs, embeddings)
```

Retriever

```
In [12]: retriever = db.as_retriever()
```

```
In [13]: query = "AWS S3에서 데이터를 불러올 수 있는 DocumentLoader가 있나요?"
```

```
context_docs = retriever.invoke(query)
print(f"len = {len(context_docs)}")
```

```
first_doc = context_docs[0]
print(f"metadata = {first_doc.metadata}")
print(first_doc.page_content)
```

```
len = 4
metadata = {'file_type': '.md', 'file_name': 'README.md', 'source': '.devcontainer\\README.md', 'file_path': '.devcontainer\\README.md'}
```

Then you will have a local cloned repo where you can contribute and then create pull requests.

If you already have VS Code and Docker installed, you can use the button above to get started. This will use VSCode to automatically install the Dev Containers extension if needed, clone the source code into a container volume, and spin up a dev container for use.

Alternatively you can also follow these steps to open this repo in a container using the VS Code Dev Containers extension:

1. If this is your first time using a development container, please ensure your system meets the pre-reqs (i.e. have Docker installed) in the [getting started steps](<https://aka.ms/vscode-remote/containers/getting-started>).

2. Open a locally cloned copy of the code:

RetrievalQA (Chain)

```
In [14]: from langchain_classic.chains import RetrievalQA
        from langchain_openai import ChatOpenAI

        llm = ChatOpenAI(model_name="gpt-4.1-mini", temperature=0)
        qa_chain = RetrievalQA.from_chain_type(llm=llm, chain_type="stuff", retriever=retriever)

        qa_chain.invoke(query)
```

```
Out[14]: {'query': 'AWS S3에서 데이터를 불러올 수 있는 DocumentLoader가 있나요?',
          'result': '네, LangChain에는 AWS S3에서 데이터를 불러올 수 있는 DocumentLoader가 있습니다.\n\n`langchain.document_loaders.s3` 모듈 내에 `S3FileLoader` 클래스를 사용하면 S3 버킷에 저장된 파일을 로드할 수 있습니다.\n\n예를 들어, 다음과 같이 사용할 수 있습니다:\n\n```\npython\nfrom langchain.document_loaders import S3FileLoader\n\n# S3 버킷 내 파일 경로 지정 (예: `my-bucket/path/to/file.txt`)\ns3_path = "s3://my-bucket/path/to/file.txt"\nloader = S3FileLoader(s3_path)\ndocuments = loader.load()\n```\n\n이때 AWS 자격 증명은 환경 변수나 AWS CLI 설정을 통해 미리 설정되어 있어야 합니다.\n\n더 자세한 내용은 [통합 문서](https://docs.langchain.com/oss/python/integrations/providers)에서 확인할 수 있습니다.'}
```

```
In [ ]:
```